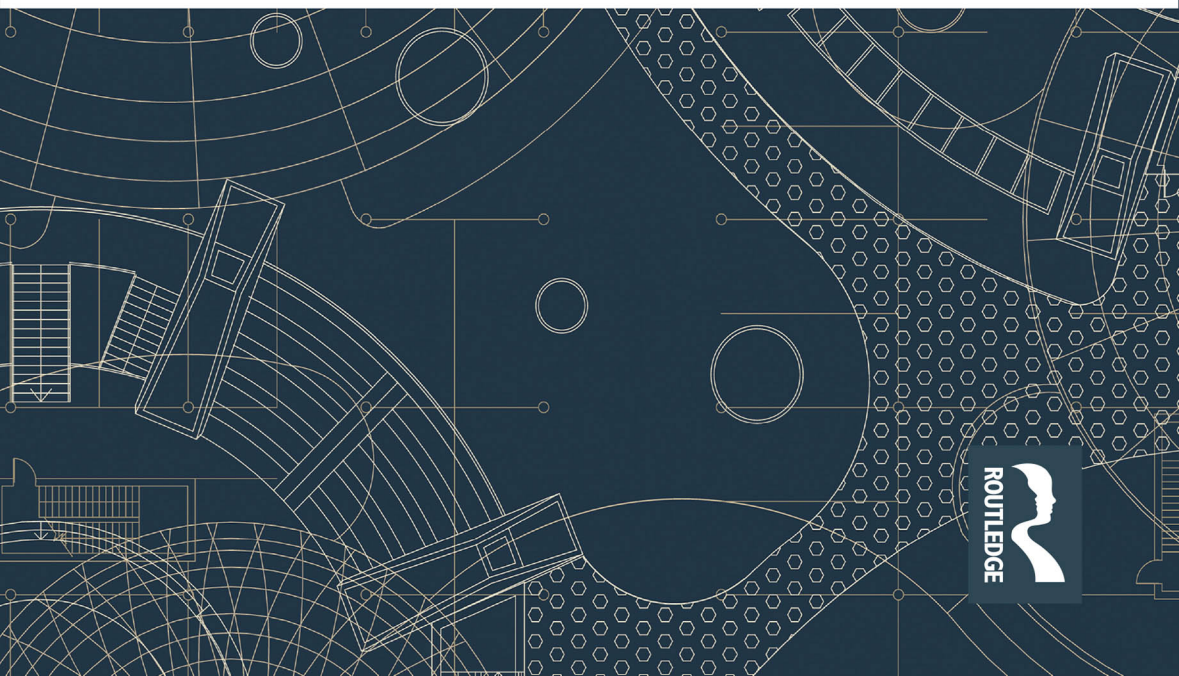




Changing Mobilities

MOBILITY INJUSTICE BY DESIGN

Ole B. Jensen



ROUTLEDGE



Mobility Injustice by Design

Mobility Injustice by Design examines the social exclusion of vulnerable people in urban spaces, revealing how millions of citizens have their 'right to roam' curbed by design and planning decisions made either intentionally to facilitate immobility or social exclusion, or because of a lack of awareness of the aggregated consequences.

Ole B. Jensen's insightful volume offers a theoretically informed framework, allowing us to 'see' mobility injustices more clearly, consider how new knowledge can mobilize design ethics, and think about how mobility justice can be brought about. It challenges us to reassess how choices of materials, design of spaces, and implementation of technologies afford or hinder particular mobilities. It explores a way of thinking about mobility and injustice that recognizes that *everyone* is assisted in their mobility practices and that human capacity for acting in the world is mediated by spaces, things, artifacts, and technologies. It examines not only the centrality of design but also the possibility that design may be working counter to the widespread cultural belief that design is in the service of 'the good'. It locates thinking about mobility injustice between affect, experiences, and sensations on the one hand and reason, principles, and formal rules on the other. Focusing on three realms of practice, it reveals the ways in which the design of cities and urban spaces accommodates the needs of people living as unhoused, disabled, and aged and how these groups experience mobility injustice. Crucially, it provides insight into the potential to move toward a future, more mobility-just society on a local, national, and global scale, in light of warfare, environmental, and migration crises.

Mobility Injustice by Design informs the future agenda of both critical mobilities research and the design fields and reflects on the potential for those in the arts and design to provoke public engagement in these timely issues. It is essential reading for all those concerned with the study and practice of mobilities theory, urban sociology, urban planning and design, architecture, urban geography, disability studies, and gerontology.

Ole B. Jensen is Professor of Urban Theory at the Department of Architecture, Design and Media Technology, Aalborg University, Denmark. He is the Deputy Director and Co-Founder of the *Centre for Mobilities and Urban Studies* (C-MUS). He is the author of titles including *Mobilities Design*, *Urban Designs for Mobile Situations*, and *Staging Mobilities*.

Changing Mobilities

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Mobility Injustice by Design

Ole B. Jensen

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Preface

This book is special. Perhaps all books are, but this one is truly special. It represents the coming together of several ideas and thoughts collected over many years of research into cities and mobilities. However, it also represents my return to questions of power and exclusion – themes that set me in motion as a young sociologist trying to understand the world of urban planning and design back in the mid-1990s. For reasons too numerous to mention, that focus was not always there as a guiding principle. But now it is. Hence, the ‘return’.

No book is a result of an individual author’s work alone, and neither is this one. I want to thank several people and institutions. Moving by the latter first, I want to thank *the Department of Architecture, Design and Media Technology* (CREATE) at Aalborg University for hosting me as Professor of Urban Design and Urban Theory. This year, I have been a Full Professor for 20 years. It has been a true privilege to learn from design and architecture colleagues and students along the road. I want to thank the members of the *Urban Design* research group for great conversations and feedback over the years. Nested in the Department but breaching out beyond its boundaries, *the Center for Mobilities and Urban Studies* (C-MUS), which I co-founded in 2008, should also be mentioned. We modeled C-MUS after our ‘sister’, the *Center for Mobilities Research* (CeMoRe) in Lancaster, UK, founded by my late and highly missed colleague and mentor, John Urry. C-MUS has proven to be a perfect sand pit for pitching loose ideas, halfway-written papers, and thoughts for research applications and projects. Especially, I want to thank the many students and staff attending our ‘Brown Bag’ research seminars over the years.

The Velux Foundation with its ‘Hum Praxis’ call has supported the research into Exclusionary Design with a generous grant that allowed for a PhD stipend that I had the privilege to supervise. The project ran over a four-year period (2021–2024), researching how unhoused people experience social exclusion on the streets of Copenhagen. In the research project ‘Exclusionary Design – social exclusion in public spaces’, I had the opportunity to collaborate with some great folks and minds. Thanks to Pia Justesen, my ‘partner in crime’, for making the application come alive in the first place, and for your contribution to bringing human rights and law into the discussion (and my awareness). Also, thanks to PhD student Carsten Hvid Nielsen for taking on the task of exploring the everyday life of unhoused people in

Copenhagen. The fieldwork was challenging for you and very enriching for me to discuss. In the project, I want to thank Bo Heide Jochimsen from the NGO 'Projekt Udenfor', Ask Svejstrup from SAND, Graham Yantis from Kirkens Korshær, Pedro Casado Michael from Copenhagen Municipality, and Michael Lodberg Olsen from MINORITET. They were all part of the steering group of the research project and have shaped and formed my understanding of the conditions for the unhoused during these years of collaboration. From Projekt Udenfor, Kirsten Skovlund Asmussen and Anne-Sofie Maini-Thorsen should be thanked for their constructive and stimulating collaboration in relation to setting up the exhibition 'Den Udstødende By' (The Exclusionary City), disseminating key findings from the research project on 'Exclusionary Design' in Copenhagen, October 2024.

Associate Professor Robert Rosenberger, Georgia Tech, USA, should be thanked not only for being such a perfect host to the PhD student from the research project during his 'stay abroad' but also for proving to be a kindred spirit with an interest in exclusionary design and post-phenomenology – two themes we have not exhausted the potential of yet.

The *Bevica Foundation* should be thanked not only for granting me a Bevica Foundation Travel Scholarship and allowing me to do some of the field work in Japan that this book draws upon, but also for introducing me to the ideas of 'Universal Design' and for letting me see the commitment and engagement the Universal Design Hub tirelessly lays out in order to 'Leave no one behind'. Thanks to Camilla Ryhl, Marie Engberg Eiriksson, Stine Louring Nielsen, and René Sørensen Overby for inspiring conversations and support. Lastly, I also appreciate having been invited into the *Bevica Research Network*.

I want to thank Pete Adey and Monica Büscher, who, as editors of the series *Changing Mobilities*, took on yet another manuscript from my hands. On a technical note, it should be mentioned that all interviewees' identities are concealed. Any name figuring in relation to a respondent's statement is fictitious. When there are quotes and statements of an 'empirical nature', these are set in italics. The sources are a mix of primary data from the Exclusionary Design research project and secondary data sources.

In terms of the terminologies used in this book, I have tried to land on an inclusive and respectful tone. However, since the communities of people in homelessness, senior citizens, and humans with cognitive and physical challenges are very diverse and heterogeneous, a unified and consensual nomenclature seems out of reach. Hence, some prefer the first-person language (e.g., 'people experiencing homelessness' or 'people with disabilities'). Here, I have chosen the terms (that also have their opponents and proponents) of 'unhoused', 'aged', and 'disabled'. In a similar vein, I have opted for the term 'street sleeper' even though some would rather use the term 'rough sleeper' (here might also be a UK/US English distinction at play). However, the general point I wish to make is that I recognize the importance of utilizing respectful and inclusive language, even if at times it can be challenging to settle on the choice of terms.

Thanks to student assistant Andrea Bruun Jensen for organizing the systematic literature search within the areas of mobilities, aging, homelessness, and disability.

Thanks to Anders Kristian Sanderbo who made the eminent sketches to accompany the dry prose of this book and thus brought the mobile situations to life via his amazing illustrations.

This work is nested within the global mobilities research community, and many have influenced my thoughts and work. I want to extend a special thanks to my friends and colleagues Mimi Sheller, Malene Freudendal-Pedersen, Sven Kesselring, Phillip Vannini, Kaya Berry, Martin Tranberg Jensen, and Chris Schimkowsky for being such inspiring colleagues, constantly reminding me about the relevance of mobilities research. Special thanks to my ‘critical friends’ who read the manuscript and commented in the 11th hour: Pia Justesen, Malene Freudendal-Pedersen, Chris Schimkowsky, and Carsten Hvid Nielsen.

Even hard-working professors have a life outside academia. Playing my electric guitars has been an integral part of me since the age of 10 and I cannot imagine life without playing music. So, thanks to the many different playmates who have endured my playing and company across genres and venues too many to explain. I enjoy playing in all of them. Expressing yourself in tones is quite a magnificent thing!

Closer to home and heart, my late father and my younger brother both showed me that we all live ‘different lives’ and that living differently has its own values and virtues, as well as challenges and troubles. Last, but not least, thanks to my close family who are not only always there as support but also the reason all this stuff is worthwhile doing. Doing whatever one can to make this troubled world a better place often seems futile. Lone, Christine, and Julie, you make the whole venture make sense to me.

As is obligatory in a preface the last sentence reads: even though many were called upon, the responsibility for any faults or mistakes is only to place upon yours truly.

Ole B. Jensen
Aalborg
1 November 2024



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Part I

Positioning



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1 Introduction

The city is not a container where differences encounter each other; the city generates differences and assembles identities ... the city is a battleground through which groups define their identity, stake their claims, wage their battles and articulates citizenship rights, obligations and principles.... The city as a difference machine relentlessly provokes, differentiates, positions, mobilizes, immobilizes, oppresses, liberates. Being political means being of the city. There is no political being outside the machine.

(Isin 2002:26)

Try to imagine that you are walking around in the city after dark, looking for a place to sleep on the streets. This is reality for millions of unhoused and street sleepers on city streets around the world every day. Often, they will be met by benches angled so horizontally placed bodies will fall off simply due to gravity, or pointy metal spikes inserted into surfaces under bridges, where one could have had shelter. Or you might be a senior citizen struggling to walk the rumbling stone pavements of the city streets. Worse it will only get when you are trying to cross the big street with light crossings. The ‘green period’ that is designed into the system has its priority to make cars circulate the city as seamlessly as possible, with the result that slow-walking citizens cannot cross roads quickly enough. As you start experiencing this over time, you tend to become less interested in going out into the city and you may stop doing so completely. Or you might be a wheelchair user struggling to move around in the city. High curbs, lack of ramps, and artifacts such as cafe chairs or advertisement signs on the sidewalks make the daily urban trip a hassle rather than a joy.

According to the United Nations, the number of people aged over 65 is projected to double from 761 million in 2021 to 1.6 billion in 2050 (UN 2023). Simultaneously, it is estimated by the WHO that 1.3 billion people experience a significant disability. This represents 16% of the world’s population – or 1 in every 6 (WHO 2023). In the EU, 20% of all citizens were over 65 in 2020 (Eurostat 2020), and over 100 million citizens have some form of disability (European Commission 2024). Despite vast inaccuracies in measurement, it is estimated that close to 895.000 humans live in homelessness in Europe (Feantsa 2023). Globally, it is estimated that at least 150 million, or approximately 2% of the world’s population, are unhoused (WEF 2021). Trends are highly uneven across countries, but a

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general decrease in the number of humans without adequate shelter is not on the horizon (Chamie 2020). Despite these numbers, cities are still mainly built for the able-bodied citizens (Falster 2024; Gissen 2022), nested in middle-class values, and for ‘ordinary’ citizens (e.g., not people in homelessness) (Jensen 2022; Nielsen 2024a). This touches on the fundamental question: “Who is the built world built for?” (Hendren 2020).

This chapter presents the key research questions and how the matters of concern are related to wider issues inside and outside of academic research. The motivation of the research into mobility injustice by design is laid out, and the emerging research field of mobility injustice and its relation to design is presented. From the mobility patterns of the everyday life movement on busy city streets to the global refugee crisis, we see mobility immanently related to questions of injustice. This chapter takes this basic insight and connects it to the question of design and ‘making’. Illustrating that the ‘artificial’ nature of mobilities and the fact that it is designed (‘made’) means that it can be different and that it can be challenged and changed. This chapter presents the general idea of this book and its content and structure.

To move is a fundamental activity for all life forms, humans included. What enables movement and mobility differs greatly from place to place, from individual to individual. As a human, the crucial moment when one stands upright and takes the first steps into the world for the very first time is not to be underestimated (just ask any proud parent). To move by one’s own bodily will and force, and assisted by one’s cognitive and mental skills, is so profoundly intertwined with ‘becoming human’ that we perhaps take it for granted. Equally, is the autonomy experienced when riding one’s bike for the first time, or when a driver’s license enables the first car ride, often understood as central hallmark of autonomy and self-determination. The ‘right to roam’ has become just that – a right! In the United Nations’ Universal Declaration of Human Rights, mobility is one right among all the others (UN Human Rights, Article 13). At least this autonomous, self-moving subject is a key figure in much of politics and mobilities thinking. However, such an idea of an autonomous mobile subject is biased. Mobilities is much more differentiated and, for instance, mediated by capacities and resources (e.g., abilities) but also relationally dependent on others and thus related to caring relations (e.g., the carrying of infants’ bodies before they can move as well as various forms of assisted mobilities). The care of other people to assist in making mobility means to become a ‘mobile with’ (Jensen 2013). The assistance of artifacts, technologies, or other bodies illustrates that the figure of the autonomous mobile subject is fictional. Even so, the trope is powerful and has found its way into foundational policy documents and generic ideas of what it means to be human.

Regardless of whether we rely on others or not, and regardless of the degree of reliance on technologies and artifacts, we could (or should) react when our ability to move is curbed and restricted by bodily, cognitive, social, or other forms of restrictions. There are important issues to be understood when someone’s mobility becomes the immobility of others. This is, for example, the case when big road infrastructures create barriers between local communities, or when digital

technology requirements favor some citizens' mobility relative to others. Also, the design of technologies, vehicles, infrastructures, and spaces may become complacent and restrictive in terms of how we may enact our rightful mobility. This work is growing out of the situational mobilities research asking questions such as 'how does it feel to move/be moved?' (Jensen 2013) and it connects these to the perhaps most fundamental one: 'Who is this world designed for?' (Hendren 2020:10). In combination, these two simple (but yet complex) questions span out a research space to be explored, and where this work is only one smaller contribution.

This book explores the relationship between the mobilities being curbed by design decisions and their material conditions. It asks questions about how choices of materials, design of spaces, and implementation of technologies afford or hinder mobilities. Hence, it connects to the emerging research agenda of 'mobility justice' (Sheller 2018), 'design justice' (Costanza-Chock 2020), and 'design ethics' (Wakkary 2021). It engages the artificial, designed, and 'made' human habitats of mobilities with critical questions related to: who counts as passengers? Who has the right to move? Who has access? Who is prevented from moving? All with the intent to show how informed design choices are shaping the material conditions of humans' lives in diverse areas such as street sleepers, disabled and impaired people, and aging citizens. Across the three areas of *unhoused*, *disabilities*, and *aging*, this book shows how mobility justice is at risk. My research in 'exclusionary design' (social exclusion of vulnerable people in urban space) is the outset (Edin 2017; Jensen 2019), but the mobility injustice questions related to this research are broadened out to also include disabilities and aging.

Mobility Injustice by Design draws references to the fields of critical disabilities studies, design justice, and mobility justice, and teases out lines of enquiry from these to a wider set of questions of concern to all design alert citizens: How does one 'learn to see' the normativity of design? How are values inscribed into 'made' and designed spaces, artifacts, and technologies? How do we orchestrate an informed conversation about 'commoning' the values and normativity of design? How are these lessons valuable for questions of transformation toward more just and sustainable social futures? How is design excluding unhoused street sleepers, disabled, and senior citizens from moving freely within the infrastructural landscapes of contemporary cities? Can mobility justice be brought about, and how can new knowledge mobilize a design ethics aiming at this matter of concern? These are the key issues of this book.

Cities – sites of coexistence and exclusion

From Aristotle (1995) and onward, the city has been seen as deeply intertwined with issues of power. Here, I can only insert a few needle pins into the vast history of urban theory. One place to start is the French urbanist and philosopher Henri Lefebvre and his pertinent question about who has the 'right to the city?' (1991, 1996, 2003). The social processes shaping cities, and their effects on humans, were a key concern for Lefebvre, who argued that to 'say "urban space" is to say centre and centrality' (1991:101). Nested into regional, national, and global processes of

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production and consumption, cities became central places for commerce, culture, and power (Soja 1989, 1996, 2000). Also, the work of Marxist geographer David Harvey is central. The role of Capital in the history of urbanization is a pivotal theme but also the more recent developments of gentrification and the rise of new types of cultural classes in the city that drives the ‘upgrading’ of spaces and the conflict over access to urban space (1989, 1990, 1996). Cities are networks of production and consumption, and the history of urban transformation under Capital is indeed the history of technological development (Castells 1996, 1997, 1998; Susser 2002). In the words of Harvey,

The more global interrelations become, the more internationalized our dinner ingredients and our money flows, and the more spatial barriers disintegrate, so more rather than less of the world’s population cling to place and neighborhood or to nation, region, ethnic grouping, or religious belief as specific marks of identity.

(Harvey 1996:246)

Loosening our grip a bit from the Marxist analytical framework, urbanist Richard Sennett argues that cities are both material artifacts and social processes. Or as he puts it: ‘urbs’ and ‘civitas’ (Sennett 1990). The former means stone and is recognized in the term urban or urbanization. The latter connotes civilization and is known in the term civility. However, despite its more or less justified role in the development of (Western) civilization, cities are also sites of conflict over habitation space, and this is in particular visible when it comes to public spaces (Lofland 1998). Complex processes of capital accumulation, gentrification, and the quest for power have left urban spaces in most cities as proverbial ‘battlefields’ for social groups and interests (Davis 1992; Graham 2010; Simpson et al. 2017). In their agenda-setting ‘Splintering Urbanism’ from 2001, Graham and Marvin argue that infrastructure, technology, power, and social stratification are woven together in disorganized and contested spaces. Or, to return to Harvey, who argued

Places ... become much more concerned about their ‘good business climate’ and inter-place competition for development becomes much more fine-tuned. The image-building of community (of the sorts which characterize Baltimore’s inner harbour) becomes imbedded in powerful processes of inter-urban competition.

(Harvey 1996:247)

Even though historical analysis of urban publicness has had a tendency to locate ‘The People’ in symbolically significant plazas and central squares, the contemporary and ‘splintered’ city has become a patchwork of emerging ‘urban domains’ (Hajer & Reijndorp 2001) rather than a central and fixed ‘agora’ (Jensen 2020b). In the early Chicago School sociology (Wirth 1938/96), Simmel’s idea that the city is a site of exchange among strangers (1903/95, 1908/50) grew into the understanding

that cities have the potential to foster peaceful coexistence and tolerance (the ‘civitas’ in Sennett’s understanding):

The juxtaposition of divergent personalities and modes of life tends to produce a relativistic perspective and a sense of toleration of differences which may be considered as prerequisites or rationality and which lead toward the secularization of life.

(Wirth 1938/96:193)

The ‘politics of difference’ (Young 1990) was therefore on the one hand seen as the promise of urbanism. The same idea was found in urbanist Jane Jacobs’ analysis of cities. She argued that multiple cultures and identities were the creativity- and productivity-enhancing competitive advantage of cities (Jacobs 1961). In a similar vein, Sennett argued for pluralism and difference:

We need to see differences on the streets or in other people neither as threats nor as sentimental invitations, rather as necessary visions. They are necessary for us to learn how to navigate life with balance, both individually and collectively.

(Sennett 1990:xiii)

On the other hand, the dense inner urban city spaces that need to be shared among a plurality of social groups with divergent interests spell conflict and territorial controversies (Davis 1992; Graham 2010). The history of urbanization and its mirroring in urban theory, sociology, urban geography, architecture, and many other strands of theory is too vast a topic to cover in detail here. What I wanted to suggest is that the city as a site of cohabitation as well as conflict has been the ground condition for long. Contemporary development of capitalism, globalization, and technological development has given new shapes to urban life, and these will be drawn out in the chapters to follow. For now, understanding the city as a site of great promise and despair that affects the lives of unhoused, disabled, and aging people is the key.

Mobility

This work is part of the so-called ‘mobilities turn’. Most often identified via the landmark text *Sociology Beyond Societies – Mobilities for the 21st Century* by the UK sociologist John Urry published in 2000. The field is wider and contains multiple references and resources that will be drawn into the arguments along the lines of this book. There are many overview texts and reference works that also may serve for identifying the ‘turn’ (Adey et al. 2014; Jensen 2015; Sheller 2021). For simplicity and clarity, I will not lose myself in the genealogy of the field from this outset but simply try to convey a few points of relevance from the ‘mobilities turn’. The first thing to notice is that social science in general took a ‘spatial turn’ in the 1980s. Most well known perhaps by the work of the UK sociologist Anthony

Giddens and what has been considered his main theoretical work *The Constitution of Society* (1984). The spatial turn was a progressive engagement with key insights from human geography and other ‘spatial’ disciplines. Mainstream social science was criticized for being ‘blind’ to the importance of spatiality to human action and social relations.

The second turn might then be said to occur with the ‘new mobilities turn’ from around the millennium. The insights from the spatial turn had sunk into many areas of social thinking. However, the meaning of mobility and the fluid network character rather than a sedentary fixed societal structure led scholars like Urry (2000), Cresswell (2006), and Sheller (2021) to problematize the static and sedentary thinking leading to seeing ‘society as a thing’. Rather, it was to be understood as a set of relations (something already Simmel noticed well before, Jensen 2006). Moreover, if ‘the social’ is a web of relations, then the movement of messages, people, goods, and ideas becomes key. Here was the second dimension of the critique that the mobilities turn carried with it. Instead of thinking of mobility as the instrumental displacement of things, objects, bodies, and ideas in space from point A to point B, the mobilities turn argued that mobilities is ‘more than A to B’. The approach wanted to unlock the understanding of mobility simply as transport (or accessibility) to unravel the more fine-grained dimensions. One way of arguing about the ‘added value’ by applying the mobilities perspective was to say that mobility (or immobility) transforms the ways we think about the world, the ways in which we relate to other people, and the ways we think about ourselves. Spoken simply, mobility reconfigures materiality, sociality, and identity (Jensen 2013). So, the turn to mobilities in social theory enabled an understanding that commuting or lack of access to buildings was about much more than simply movement. Such sociological and anthropological insights became incorporated into the turn and worked as a powerful critique of mainstream sedentary social thinking as well as instrumental understandings of transportation and accessibility.

Within the last decade, the mobilities turn (which itself is a very hospitable and open field) has taken yet another turn, namely one toward design. Over that period, the articulation of ‘mobilities design’ has become one important sub-set to the mobilities turn (Jensen 2014, 2023b; Jensen & Lanng 2017). The research in mobilities design grew out of an interaction with urban design and architecture. Two main things were illustrative of this interaction. The first one was that by engaging with urban design and architecture, a whole new, spatially sensitive vocabulary became available. Terms like *volume*, *surface*, *materials*, *atmosphere*, *light*, *sound*, *microclimate*, and *shade* entered the vocabulary of the mobilities design research field (Jensen 2023b). Second was the critical-creative thinking, or what has been termed ‘designerly ways of thinking’ (Jensen & Lanng 2017). As those who work with designers will know, the outset is rarely a problem or a research question, but rather a critical-creative and highly explorative ‘what if?’. As the social sciences have developed a practice of working with ‘problems’, the designers seem to engage more with ‘potentials’ (Jensen 2024a). The research into mobilities design is thus opening toward a more granular understanding of the material world, as well as the innovative and explorative processes of imagining

future transformation and change. In this way, the mobilities research became materially and creatively better attuned.

However, mobility is a social phenomenon that begs classic social science awareness of power and justice. Mobility is a stratifying factor. Since our birth, we roam the world as expressions of our autonomy. Never unhindered of course. Often inhibited by rules and formalities, by powers of paternalism (parenthood has much to do with putting curbs on mobility for the child), our own ‘naked capacities’ (Ihde 1990), our cognitive ditto, the systems, infrastructures, spaces, and technologies ready at hand for assisting and affording mobility. So, as individuals, we recognize mobility as a right and resource. This is why it is in the Human Rights Declaration and in many other documents. At the same time, as mobility is tightly connected to autonomy and freedom, it is connected to power, injustice, and exclusion. Zygmunt Bauman argues that ‘capacity to move has become the major, perhaps the paramount stratifying factor of the emergent global hierarchy’ (Bauman 2002:83). As the nonbinary trans feminine scholar Sasha Costanza-Chock argues: ‘I walk through a world that has in many ways been designed to deny the possibility of my existence’ (Costanza-Chock 2020:5). Costanza-Chock narrates the experience of being subject to the airport security scanning that has been programmed with a gender binary perspective as one of many experiences with exclusion being ‘designed in’ to the infrastructural landscapes of the everyday life. We are ‘over a life course ... all to some degree prosthetic citizens whose capacities for mobility depend on the constraints and affordances of the public sphere’ writes Sheller (2018:55).

In general terms, this book has its outset in the mobilities research understanding. However, to qualify and target the scoping relevant to exclusion of unhoused, disabled, and aging, I want to supplement with a more detailed framework focusing on bodies, design, and injustice. The following three sections lay out some of this territory.

Bodies

In light of the theoretical position taken in this work, I explore a way of thinking about mobility and injustice as related to the fundamental fact that everyone is assisted in their mobility practices. Able-bodied might have a different kind of assistance, but nevertheless so. This perspective draws on the idea that human agency and capacity for acting in the world (including moving) are always mediated by spaces, things, artifacts, and technologies. Inspired by the post-phenomenology of Don Ihde, we might say that rarely are we relying solely on our ‘naked capacities’ (Ihde 1990). Rather, we should think of our actions in the world as mediated, assisted, and distributed. Elsewhere, I have argued that we have ‘extended bodies’ and act in ‘elastic situations’ (Jensen 2021). As, for example, seen in an analysis of the regulation of public space behavior under COVID-19,

Our bodies connect, relate, sense, and touch in various ways upon and mediated by different forms of materials (from air and particles to hard surfaces and human-made artefacts). The notion of the ‘extended body’ is one way of articulating this assemblage of body and context.... Our practical and emotional

reactions to the calculative and governmental articulation of ‘safe social distances’ [during Covid-19] makes visible the fact that we already have norms and codes for this in the everyday. The subtle calibrations of the ‘proper’ interactional distances are being exposed in the interventions and safety protocols related to Covid-19. Looking at these from the point of view of the ‘elastic situation’ renders the authority’s policies more than simply governmental procedures. They become ‘experiments’ of the situational dynamics.

(Jensen 2021:5–6)

Within these elastic situations where the body’s capacity is mediated, stretched, and extended (or restricted, curbed, and obstructed), there is an important but in the wider literature on city planning and transport neglected dimension. Namely that of the embodied and multisensorial relationality with the world. We not only ‘act within’ the wider environment but also sense, perceive, and experience ‘with’ it. We are ‘augmented’ by materials when we move (or are prevented from moving):

The starting point is that the analysis of mobilities transforms the social sciences. Mobilities makes a difference ... the powers of ‘humans’ are always augmented by various material worlds, of clothing, tools, objects, paths, buildings and so on.

(Urry 2007:44–45)

Just as our capacities and agency are distributed and dependent on spaces, artifacts, and technologies so are our bodies ‘open’ toward these externalities. Elsewhere, this relationship has been described as a relation of ‘osmosis’ (Jensen 2016). Light, sound, smell, topography, shade, heat, etc. – all these elements of being ‘work on us’ in ways that often fall below the threshold of conscious awareness. When cycling, the air not only gives resistance but also mediates sounds and smells entering our bodies (Jensen 2013). When flying as a passenger in modern aircraft, pressure cabin technology and machine-mediated aerial turbulence vividly illustrate that we are ‘in’ the environment and not ‘next to it’ (Jensen & Vannini 2016). Hence, there is an ontological condition tying mobilities and bodies to the question of justice and injustice:

The problem of mobility injustice begins with our bodies, and the ways in which some bodies can move more easily through space than others, due to restrictions on mobility relating to gender, race, class, ethnicity, sexuality, and physical abilities.... Second, mobility injustice concerns the shaping of built environments by uneven infrastructures and land use, including the forms of buildings, streets. Vehicles, public transit, and other forms of transport infrastructure that have long been part of the making of racially segregated cities, sexually unsafe areas, or places of class exclusion.

(Sheller 2021:43)

Alongside such an argument for paying more attention to our ‘nestedness’ within the world, we find research into so-called ‘Skin studies’ or the ‘dermatological

turn' (Howes 2018; Lafrance 2018). Accordingly, we are related to the world via the interface of our bodies, or the 'skinscape' which is the 'contiguity or intimate association between the surface of our body and the surface of the earth, or landscape' (Howes 2018: 229). Such perspectives are found more elaborately in many non-Western Cultures. For example, among Aboriginal Australians, to whom the landscape is a 'skinscape' as 'it is composed on the material traces of the bodies or the Ancestral beings who roamed and shaped the countryside' (Howes 2018: 229). Equally, natives of Peru have 'skin knowledge' as a way to articulate the localized way-finding competencies of the Cashinahua. Marc Lafrance argues that 'far from a sealed off or seamless membrane, the skin is full of folds, pores and orifices that push it into the world and the world into it' (Lafrance 2018: 6). In other words, our osmotic openness to the world (including other bodies) is best described as a relation beyond subject and objects. Another perspective that fits this idea of the extended body is theories and research conducted on 'leaky bodies' (see Jensen 2021). Again, a perspective problematizing the Cartesian isolated subject (Grunsvén 2024).

From the generic insight that all bodies are relationally and interdependently entangled with their environments, it becomes important to notice when different bodies have different needs and different capacities. Needless to say, one should be careful to naturalize the differences, but the opposite standpoint (that people with special needs have precisely 'special' needs and hence need 'special' treatment) may indeed also be unhelpful. To be identified as 'special' may be part of a process of stigmatization (Goffman 1964). Moreover, it may point toward 'special' solutions that, for example, are not on the public budget or that need separate political negotiation and attention. Here, I am leaning more toward some of the ideas underpinning the movement of 'Universal Design'. At least, the idea that spaces and environments should be designed so that they are useful for everyone and not some target group or people with special abilities (e.g., Bjerke 2023; Frandsen et al. 2023; Iwarsson & Ståhl 2003; Lid 2014; Mace 1985; Ryhl 2018; Steinfeld & Maisel 2012; Story 2010; Story et al. 1998; Strandgaard 2018; Strandgaard & Lygum 2022). Even though this sounds compelling, the question of deciding 'who counts' and how 'for all' is identified in practical terms seems complicated (Hamraie 2017; Sheller 2018).

This work is inspired by the critical insights coming from the mentioned Universal Design perspective. However, it will pay homage to the eclectic culture of the mobilities research and reach out and include other perspectives as well. Here, I am among others thinking of the contemporary discussion of 'design justice' (Costanza-Chock 2020), 'design ethics' (Wakkary 2021), 'mobility justice' (Cook & Butz 2019; Sheller 2018), 'age-inclusive public spaces' (Hauderowicz & Serena 2020), a 'cultural politics of emotion' (Ahmed 2014), a 'performative sensitivity to the assembling of vulnerable bodies' (Butler 2020), 'pluriverse thinking' (Escobar 2018), an 'ethics of care' (Held 2005; Tronto 2020), 'design emergency' (Rawsthorn & Antonelli 2022) and 'critical mobilities thinking' (Jensen 2023; Jensen & Lanng 2017). Furthermore, the work is connecting to a horizon of disability studies (e.g., Goodley 2017; Furu 2022; Shakespeare 2018) and a perspective seeing design

as immanently political, normative, and culture-led (Goh et al. 2022; Kries et al. 2018; Norman 2013; Papanek 1971/2019).

Bodies was the first of the three additional frameworks that I propose. The second is related to the notion and concept of design, so now I will move to this.

Design

The whole foundation for the mobility practices (just or unjust) is the material environments and physical spaces and sites within which the mobility takes place. As Coccia argues, there are no ‘natural’ environments (2021). In a similar vein, Latour argues that to be human is to be in an assisted or artificial ecology. Latour terms these ‘envelopes’ (2008). We are living in artifacts and thereby in environs that have been designed. What that means is that design is both complex and contested. So, I will as an outset present a quick glance over some definitions (list made on the basis of Jensen 2013:177):

- Design reaches back to the Latin word ‘designare’ which means to ‘mark out or designate’ (Webster’s Dictionary).
- Design is a verb implying ‘to set something apart for someone, to intend, to make an imaginary sketch’ (Oxford Dictionary).
- Design means the deliberate shaping of the environment in ways that satisfy individual and societal needs (Donald Norman).
- Design is the conscious process to develop physical objects with functional, ergonomic, economic, and aesthetic concerns (Rune Monö).
- Designing means devising a form for an object without having that actual object in front of you (Christian Gänschert).

Also, design is at times seen as a mix between something rational and calculable on the one hand, and then the creative and artful on the other hand (Lynch 1981:290). The nature of the relationship between the designed spaces and human actions is a complex one. In human geography, sociology, and urban studies, we at times speak of a socio-spatial relation or dialectics (Tonboe 1993). Clearly, the designed spaces and systems cannot determine human behavior and practice. On the other hand, it seems naïve to ignore that they play a significant role in making human actions possible at all. So, there is a subtle relation at play here (and one I will return to many times during the course of this book). The old statement by Winston Churchill may be referenced here. As the story goes, Churchill was trying to raise money for the rebuilding of the English parliament after its damage by German bombings during WWII. In his speech, he famously stated that ‘first we create the buildings, then they create us’. The key point here is the complex, intertwined, and circular relationship that exists between us as human bodies and the material environs wherein we exist. As already mentioned under the section about bodies, this means that we are enrolled and related, not separate from our environment. However, it also means that someone somewhere has decided how these habitats and environs should materialize and manifest themselves. They are

results of design decisions and design practices. Therefore, it becomes central to the aim of this book to understand not only what design is and what it does, but also that design may be working counter to the widespread cultural belief that design is in the service of ‘the good’. Design is not necessarily liberating and emancipating. Rather, and this is one of the key insights behind this book, design may work to exclude and to ‘tell off’ humans from various activities and places (Jensen 2019, 2020, 2023a, 2023c).

So, what kind of design am I interested in here? Working on the assumptions of a ‘situational pragmatism’ (Jensen 2013), the answer to this question is that whatever design is relevant to the mobile situations I am exploring. Hence, at times, it is the choices of surface materials (e.g., asphalt or cobblestones); at other times, it is the algorithm that provides cars with longer periods of ‘green’ than pedestrians. I am, in other words, not ‘only’ interested in architecture or in urban planning typologies. Rather, I am interested in what enables or prevents the mobile situations that I explore. This is why I speak of ‘mobilities design’ as the field exploring the design choices and interventions enabling or preventing specific modes of mobility or different ways of moving. I am studying urban designs for mobile situations (Jensen & Lanng 2017) but in its widest possible sense. I am not, then, confined to a strict disciplinary understanding of, for example, urban design as being the layout of public space (which I am also interested in). Rather, I subscribe to more radical and interdisciplinary perspectives on design. One example of such a perspective is Yaneva’s understanding of design as ‘enacting the social’:

Design [is] a way of producing additional attachments that make a variety of actors congregate, forming different groupings and assembling social diversity. Tracing networks with wood, steel, polished surfaces and blinking signals, bip-bing doors and blinking elevator buttons, design connects us differently, linking disparate heterogeneous elements and effects, thus entering a game of producing, adjusting and enacting the social.

(Yaneva 2009:282)

‘User group targeting’ is one of the core disciplines in any kind of design; who are the imagined future users of the technologies, spaces, and systems that are being designed? In relation to this book’s ambition of engaging with vulnerable users, the general idea of designing for an ‘average’ or a ‘normal’ human subject is deeply problematic. Critical disabilities research has illustrated this forcefully (Gissen 2023; Hamraie 2017), and I return to this in all three chapters on the empirical material. In the practice of ‘Universal Design’, the imaginary user is also rendered an open and nonrestrictive subject. How this is manifested in practical terms must be decided in the specific situation, but the ambition is to make spaces, artifacts, and environs that are useful and open to ‘all’ (Mace 1985; Ryhl 2018). As a small side remark, one might notice that exploring injustice as an effect of design is a perspective explored before. As, for example, in the rehabilitation psychologist, Carolyn Vash’s conference presentation titled ‘Discrimination by Design: Mobility Barriers’ from 1972 (Hamraie 2017:139).

From bodies and design, I now turn to the third analytical ‘prism’ of the framework, which is injustice.

Injustice

Justice and questions of injustice might be framed philosophically and abstract, but they are very concrete. Our experiences with, for example, being jumped when standing in line are very practical (Davidson 2023:1), and so is the feeling of not being heard when someone overlooks our presence in a social situation. Various theories and philosophical positions might help us distinguish between different manifestations and gestations of justice and injustice, but at the end of the day, justice is a very concrete thing according to Davidson. Inspired by pragmatism, the question of justice is related to ‘figuring out what works’ (p. 249) and doing so in a social and collective setting. This work will lean heavily on this assumption and hence also constantly try to be empirically operational rather than abstract philosophical. Drawing a deep line back into the genealogy of justice thinking, Davidson points at a constant balancing act between justice as feeling and thinking (p. 6). Spoken in more contemporary terms, I locate the thinking about mobilities injustice between affect, experiences, and sensations on the one hand and reason, principles, and formal rules on the other hand. Theories of justice, hence, tend to fall into two main areas: one focusing on welfare and one focusing on freedom (Davidson 2023:21). According to Barnett,

... normative understandings of justice are worldly principles, which emerge from situated conflicts and orient action ... critical analysis starts not so much from a clear-sighted definition of justice but from a shared abhorrence at some form of wrong.

(Barnett 2017:241)

To ‘sense some form or wrong’ is a foundational idea of inspiration for this book. Within transport geography and planning, the question of justice has been on the agenda for a long time (Lucas 2012; Martens 2012; Verlinghieri & Swanen 2020). More recently, the issue has been integrated into the new mobilities turn, and here the pivotal work is Sheller’s *Mobility Justice – The Politics of Movement in an Age of Extremes* (2018). It is a landmark text and sets out a critical research agenda integrating the question of justice in relation to all modes of mobilities and all scales from the body to the global. It is also a very comprehensive and globally ambitious framework. In its global ambitions, it does not go deep into more detailed areas of mobility justice and injustice. This book aims to cover just such areas: concrete, empirically, situated, and grounded cases of mobility injustice. Taken together, these works may be seen as drawing up the contours of a research field engaging mobility and injustice, in a frame that acknowledges that ‘the city is a moral universe’ (Tuan 1988). Verlinghieri and Swanen point to a useful distinction between state-centric and society-centric research on justice in transportation and mobilities (p. 2). The former has a focus on planning, governance, and regulation, whereas the latter is about culture, norms, values, and the social dynamics of mobilities. Transportation

researchers have been mostly occupied with the former, but the latter fits the new mobilities agenda very well. It shows that movement between A and B has more dimensions than the physical dislocation of bodies, goods, and vehicles. It has repercussions for notions of self and other as well as the mobility/immobility patterns that shape our understanding of the material environment. This also connects to a principal difference between transport and mobilities. Much transport research sees transport as a ‘derived demand’ (Davis 2023; Dyson & Sutherland 2021; Gilbert & Perl 2010; Metz 2016; Rodrique et al. 2006) and thus a secondary activity, whereas mobilities research sees all dimensions of movement as potentially meaningful and in need of exploration. Verlinghieri and Swanen, who come from the transportation side of research, do, however, illustrate that combinations and hybrids are emerging. One illustration of this is that they acknowledge the presence of a ‘mobile ontology’ (Sheller 2018) as a foundational idea within mobilities. They argue that

In fact, a mobile ontology becomes a core precondition for the development of a predisposition towards justice that overcomes the limits of distributive approaches, which for many critical theorists are ultimately grounded in a masculinist, Euro-American/white, ableist, heterosexual understanding of being and the wider world.

(Verlinghieri & Swanen 2020:4)

This critique is echoed in the words of Cook and Butz, who argue that

Theories of social justice are predicated on as sedentary ontology that inadequately captures a world constituted through multi-dimensional and multi-scalar mobilities enabled by material infrastructural moorings.

(Cook & Butz 2019:13)

In this work, I cannot go deep into the many different theories of justice and their philosophical underpinnings and foundations. But the critique of a primary focus on distributive approaches, as voiced by Verlinghieri and Swanen, is a core point of difference between the state-centric transport research and the society-centric mobilities research. Also, the enlargement of the question of mobilities justice from a procedural to a more cultural understanding is a marker of the mobilities turn. Nancy Cook and David Butz (2019:9) have taken on the massive task of creating an overview of the many approaches to mobilities justice. In so doing, they list several mobility justice dimensions that are relevant for an overview:

- Distributive justice
- Procedural justice
- Deliberative justice
- Restorative justice
- Epistemic justice
- Environmental/ecological justice
- Redistributive justice
- Recognition

The latter one of ‘recognition’ is important to understand in relation to the three empirical areas I cover in this book (unhoused, disabled, and aging). Besides including an understanding of how it feels to live a life in misrecognition, this dimension also touches upon the theme of affect (Anderson 2009; Bissell 2010). Something that the mobilities turn is better equipped to embrace with its sociological and anthropological understanding than mainstream transport thinking. This dimension connects to the idea that injustice is a feeling. We need to add to the notion of mobilities justice as resources and opportunities, that mobility justice is a feeling, an affect, an emotion (an idea that runs all the way back to philosopher David Hume, who believed that our conception of justice was grounded in experiences of injustice, Davidson 2023:3). Hence, I want to span out this exploration of mobility injustice in a space reaching from ‘injustice as redistribution’ and to include ‘injustice as experience’. Or, spoken on a more general note, we may think of this distinction as one of ‘injustice as access’ versus ‘injustice as affect’. Obviously, both dimensions may be present at the same time. The point here is that one might be foregrounded and the other backgrounded at specific times and in particular cases. However, and more importantly, addressing injustice as a feeling allows to include sensations, affect, and atmosphere into the account for mobility injustice. Moreover, the focus on this takes us toward the most recent development of mobilities justice thinking that connects to questions of care and an ‘ethics of care’. Partly inspired by feminist ethics of care, Verlinghieri and Swanen argue that

Adopting this ethics of care as a framework for justice implies a decisive move away from distributive or liberal approaches to justice seen as right versus wrong (Gilligan 1982; Sevenhuijsen 2003; Held 2006) towards situated discussions of how ‘mobile moral subjects’ are constituted and of the imbricated nature of movement, bodies, gender, environment, and energy ... the central question becomes how to ‘care’ optimally for the mobility needs of the most vulnerable and for the non-human environment in an ongoing, collective, and reflective process.

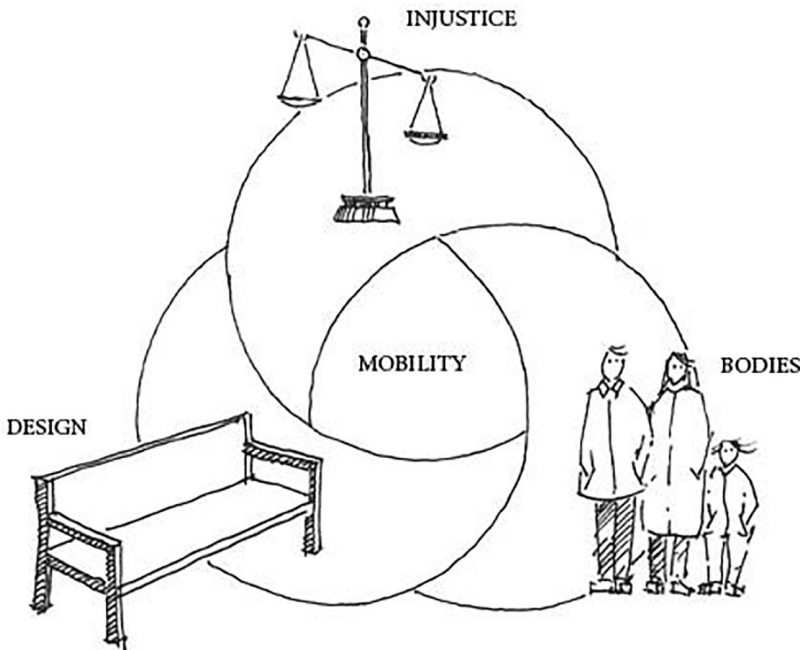
(Verlinghieri & Swanen 2020:6)

This might sound philosophically high-flying, but there are very practical and mundane consequences of this way of seeing injustice. So, I shall not ignore mobility injustices related to institutions and allocation of resources, but I will background these to focus on injustice and emotion, affect, and sensations. As I will show later, these interventions contribute to an ‘atmosphere of rejection’ (Jensen 2020). However, they are also very tangible examples of a material mediation of power. Here I am somewhat beyond (or beside) the classic concepts and theories of power. Nevertheless, it is on this level of ‘molecular politics’ (Amin & Thrift 2002) that power is exercised and injustice is materialized.

It is, of course, important to notice that some of the cases mentioned are deliberate and intentional attempts to exclude (e.g., the leaning bench working against street sleepers), whereas others are ‘unintentional consequences of design’ (e.g., the algorithms favoring fast cars over slow bodies or the lack of ramps preventing wheelchair users from free mobilities). Hence, there is a question about ‘intentionality’ to

which I return quite a lot in this work, ranging from the obvious 1:1 translation of the intention to keep street sleepers off the streets to the muddier consequences where intentions might have been good or simply overwritten by ignorance. Notions such as ‘technological intentionality’ (Ihde 1990; Verbeek 2005) are analytical attempts to nuance the simplistic understanding of intentions as something related to directed human consciousness. In line with ideas of the ‘agency of artifacts’ (Latour 1996, 2005a; Yaneva 2009, 2017), these are ways of thinking through how power and rationalities might be ‘nested’ into material choices and solutions in more subtle ways (Jensen 2019). However, if one takes a pragmatic approach, it is the felt and experienced consequences that unhoused, disabled, and seniors feel and experience that should be the point of focus and the unit of analysis. A ‘situational pragmatic’ (Jensen 2013) outset would ask ‘what enable (or prevent) this mobile situation?’ and from that seek to unravel the values, intentions, and interests in their complex intertwinements with spaces, artifacts, and technologies. This means that if you feel excluded and subject to injustice due to the material and physical layout of the city, then that is the first observation and the outset for further enquiry. Whether this is intentional or not, that will then be the next level of the analysis and exploration.

The exploration of mobility injustice by design is framed by the three intersecting themes of bodies, design, and injustice with the concrete mobile situation as the analytical nexus (see Figure 1.1).



These three key themes will be framing the analysis of mobility injustice by design and are, therefore, subsequently the focus of the three framing Chapters 3–5.

Three realms of practice – unhoused, disabled, and aged

So far, I have structured the introduction after the key framing structure of this book. Hence, I started out with mobilities, moved toward embodiment, then focused on design, and finally on injustice and power. These are the key theoretical framing themes that will be brought into play when exploring the empirical fields of mobility injustice by design. As always, one could have had a much larger sample of empirical cases, and one could have included different themes. Alas, my choice (that will be motivated along the road) fell on the three themes of the unhoused, disability, and aging. So, from here, I briefly introduce each of these.

The unhoused

The ways in which people living as unhoused and street sleepers are experiencing mobility injustice by design is the focus of Chapter 6. Based on empirical data from a research project exploring how ‘exclusionary design’ (Edin 2017; Jensen 2019) marginalizes unhoused, I connect this to secondary sources and research literature documenting the situation within this realm.

Much of the insights behind this theme is based on the research project ‘Exclusionary Design – Social Exclusion in Urban Space’ that has been conducted at the *Center for Mobilities and Urban Studies*, at Aalborg University between 2021 and 2024 (Justesen 2023; Nielsen 2024a, 2024b). The key idea has been to collaborate with the city government and NGOs in Copenhagen to see how unhoused and people living on the streets experience the increasing amount of ‘exclusionary design’ such as leaning benches, steel frames, spikes, removed urban furniture and a host of other interventions that we document not only push the unhoused away from the city but also are experienced as an ‘atmosphere of rejection’. When you are ‘living in the network’ as these people are, you do not have a home that is a sedentary and fixed place. You cannot describe it to others by reference to a fixed address or a set of GPS coordinates as most people do. Rather, you are sleeping in one place, eating in another, bathing in a third, and socializing in a fourth place. Hence, when subject to exclusionary design interventions that ‘close off’ your living room for instance the unhoused are forcefully moved around in an urban network composed of ‘go and no-go areas’ (Jensen 2019). From ethnographic field studies, the project has documented how the unhoused on the streets of Copenhagen feel mobility injustice as effect of interventions that most often are intentional and always experienced as making the city unwelcoming.

Disabilities

In Chapter 7, I explore a much larger and much less homogeneous theme of mobility injustice by design, namely the field of disability. This work is anchored in

a combination of critical disability studies, mobilities research, and Universal Design. The diversity of the field in the sense of the multiple ways in which one can be disabled makes the scoping of this chapter quite complex. However, I shall have as the focus issues and cases related to mobilities within the urban spaces of the city. But I try to cover both disabilities that are related to the usage and assistance of technologies and artifacts such as sticks and wheelchairs as well as to disabilities related to emotional and cognitive challenges that are ‘invisible’ to people not knowing the people in question (e.g., dementia, autism, or ADHD).

This chapter partly relies on the so-called ‘social model’ of disability (Goodley 2017; Hendren 2020; Shakespeare 2018). This, in short, differs from the ‘medical model’ by seeing disability as something that has to do with the relationship between the body and its environment. Where the medical model suggests that the root cause of disability is within the somatic human body, the social model is sensitive to the fact that all bodies are able at different levels and that all bodies are ‘assisted’ when it comes to being in the world. The social model of disability studies corresponds well to the ideas within the mobilities turn related to extended bodies, affordances, and the distributed agency that enables or prevents mobile situations. I also lean to some extent on the Universal Design thinking since it also corresponds with the mobilities design’s focus on the layout and design of the urban spaces of mobilities. I do, however, propose to substitute the instrumental and functional idea of ‘accessibility’ within Universal Design with the more sociologically and culturally sensitive idea of mobilities from which this work originates. The social model of disability has, however, also been criticized, and I follow some of that criticism into the field of what is termed ‘critical disability studies’ (Dokumaci 2023; Gissen 2022; Hamraie 2017).

In relation to this theme, the nomenclature is vital. In this work, I use the terms ‘disabled’ and ‘disabilities’, well-known that there are many different assessments of these terms (Dokumaci 2023; Gissen 2022; Hamraie 2017; Hendren 2020). This work acknowledges these controversies and disputes and is nested in an understanding of ‘disability’ as something relational that does not describe the ‘state of a body’ but rather the reality of a ‘disabling world’ (Hendren 2020:6). Empirically, I consult more secondary sources of data than in the previous chapter.

There is of course an overlap with the third theme since many senior citizens are suffering from different kinds of disabilities. I try to balance this as best as it makes sense in relation to the specific case and topic.

Aging

In Chapter 8, I target aging and the demographic shift that is evident all over the world today. More and more populations are growing older, and this adds renewed attention to how the design of cities and urban spaces is accommodating the needs of these groups. The focus of this chapter is the perspective laid out nicely in the publication *Age-inclusive Public Spaces* edited by Hauderowicz and Serena (2020). Inspired by this project, I will, however, focus on the injustice related specifically to mobilities in public spaces. As with the chapter on disabilities, I also draw on Universal Design

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